

Extraction Guidelines

Manually detectable Variables of Requirements Quality

This document describes how to extract all variables labeled as either “man-det” (manually & deterministic) or “man-heu” (manually & heuristic) in the [variable overview](#). The justification for their inclusion (i.e., the causal hypothesis that their values affect the relevant dependent variables) and their source can be found in the aforementioned variable overview.

Requirements-Level Variables	1
Functional Duplication	1
Use Case Naming Problem	1
Inappropriate Scope	2
Incoherent Text Order	2
Inputs or Outputs not quantitatively specified	2
Contains NFRs	3
Contains Actor-Actor Interaction	3
Contains justifications	3
Sentence-Level Variables	3
Coordination Ambiguity	3
Contains UI Design Details	4
Contains Alternative	4
Inconsistent Level of Abstraction	4
Contains Clarification	5

Requirements-Level Variables

Requirements-level variables are all variables with a scope of “Use Case” or “Requirement” and are decided per file.

Functional Duplication

deterministic	binary
----------------------	---------------

Extraction rule: Select true if the requirement describes the same functionality twice.

Examples: (not observed)

Use Case Naming Problem

heuristic	binary
------------------	---------------

Extraction rule: Select true if the requirement has a meaningless or inconsistent name. Meaningless or inconsistent names are situations where the use case’s name bears no relation with the concept described or the same name is used for different concepts

Examples:

- iTrust, UC9S1: The second sentence of the scenario goes beyond the functionality that the use case name describes (i.e., it describes a separate use case).
- iTrust, UC17: None of the scenarios describe the use case implied by the use case name. They only define some properties of patients, but no interaction that would fulfill the goal of “proactively determining needed care.”
- iTrust, UC22: The use case name is empty.

Inappropriate Scope

heuristic	binary
-----------	--------

Extraction rule: Select true if the requirement contains details irrelevant to the problem statement.

Examples:

- iTrust, UC10S1: Step 4 (graphing height or weight) is not implied by the use case name ("enter edit personal health records").

Incoherent Text Order

heuristic	binary
-----------	--------

Extraction rule: Select true if the steps of the use case in the description do not follow a logical path of events.

Examples:

- iTrust, UC3S3: The two steps of this scenario should be in the opposite order. First, an administrator should set the length of a period (currently sentence 2), then a user's session will time out when being inactive for this period of time (currently step 1).
- iTrust, UC24S1: Steps 2 and 3 represent alternatives, not consecutive steps in a procedural scenario.

Inputs or Outputs not quantitatively specified

deterministic	binary
---------------	--------

Extraction rule: Select true if any quantitative input or output is not specified with numeric values.

Examples:

- eTour, UC8: The “multiple submissions” are not quantified.

Contains NFRs

heuristic	binary
-----------	--------

Extraction rule: Select true if the requirement contains any non-functional requirements (e.g., quality requirements).

Examples:

- eTour, UC4: This use case contains a proper quality requirement appended to the use case description.
- eTour, UC27: This use case again contains a quality requirement in its description.

Contains Actor-Actor Interaction

heuristic	binary
-----------	--------

Extraction rule: Select true if the requirement contains any actor-to-actor interaction.

Examples:

- iTrust, UC29: The first step (“A patient has just been diagnosed with a condition and wants to find the LHCPs in the area who have handled that condition.”) describes the interaction between two actors (a patient and an LHCP). This is outside of the scope of a use case and rather describes a goal.
- eTour, UC7: Sentence 3 (“Check the data of the agreement and decide for activation.”) is an interaction of the user with themselves, which is technically also an actor-actor interaction.

Contains justifications

heuristic	binary
-----------	--------

Extraction rule: Select true if the sentence justifies a requirement.

Examples: (not observed)

Sentence-Level Variables

Sentence-level variables are all variables with a scope of “Sentence” and are decided per line per file.

Coordination Ambiguity

heuristic	binary
-----------	--------

Extraction rule: Select true if the sentence contains any coordinating conjunctions (and/or) that could lead to multiple potential interpretations of the sentence.

Examples:

- iTrust, UC18S2, sentence 1: The "or" is ambiguous, as the interpretation "The system shall enable the administrator to add a new entry for a hospital, [...] in an existing entry." is a second, syntactically valid interpretation of the sentence.

Contains UI Design Details

heuristic	binary
-----------	--------

Extraction rule: Select true if the sentence imposes on the solution space by constraining the UI of the solution.

Examples:

- iTrust, UC2S1, sentences 5 and 7 specify a "the hospital list pull down menu." A pull-down menu is already a design detail that should not be mentioned in the requirements. Instead, an abstract "hospital list" should have been mentioned.
- iTrust, UC22S3, sentence 5 specifies that a row is "highlighted in bold." The text formatting "bold" is a design decision that imposes on the solution space.

Contains Alternative

deterministic	binary
---------------	--------

Extraction rule: Select true if the step of the main scenario contains an alternative branch that should have been included in the extensions section of the use case.

Examples:

- eTour, UC43, sentence 4: This step contains the second sentence, "Where the data is invalid or insufficient ..." which describes an alternative path to the main scenario.
- iTrust, UC2S1, sentence 3: This step describes a main path ("The HCP saves the new lab procedure [...]") and an alternative path ("or cancels the lab procedure creation").

Inconsistent Level of Abstraction

heuristic	binary
-----------	--------

Extraction rule: Select true if the sentence is on a different level of abstraction from what a step in a scenario should be, i.e., **not** a user-system interaction.

Examples:

- eTour, UC15, sentence 6: Storing information is a system-system interaction (between the main component and a database). To the user, this interaction is not visible. Instead, the use case should specify in the postcondition that the data has changed in the system.
- iTrust, UC5S1, sentence 1: The recording of internal information is invisible to the user and only a system-system interaction.

Contains Clarification

heuristic	binary
-----------	--------

Extraction rule: Select true if the sentence contains a clarification (either of a different step or the use case in total). A clarification does not represent an interaction of an actor with the system but rather describes the properties of objects or actions.

Examples:

- iTrust, UC17S3, sentence 1: This sentence simply defines “An alive patient under the age of 19 who has not had proper immunizations per the immunization schedule.” This is not a step in a scenario, just information.
- iTrust, UC40S2, sentence 4: This sentence defines that “In the calendar, the date entry with an office visit shall display a label as the diagnoses (i.e., ICD-9CM codes) for the office visit.” This is not a proper step in a scenario (the interaction is described one step before that) but rather supplementary information.